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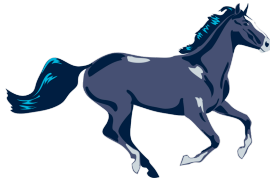
NID: A universal distance

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Problem: Can we define a concept similarity measure based on Algorithmic Information?

Complexity offers a natural measure of **concept similarity**. We may use conditional complexity $K(x|y)$ to estimate how much x differs from y .

We can turn this measure into an operational method by estimating K using Web frequencies, as we did previously.



Suppose you want to measure the conditional complexity $K(\text{rider} | \text{horse})$ between two concepts, *rider* and *horse*. You can get an estimate by computing $\log(1/f(\text{rider} | \text{horse}))$. Conditional frequency $f(\text{rider} | \text{horse})$ can be computed the same way as conditional probability: $f(\text{rider} | \text{horse}) = f(\text{rider} + \text{horse}) / f(\text{horse})$.

Let's perform some measures on a Web search engine:

- "rider" gets 74×10^6 hits on Bing,
- "horse" gets 72.5×10^6 ,
- "rider + horse" gets 13×10^6 .

The conditional frequency, according to Bing, is therefore $f(\text{rider} | \text{horse}) = 13/72.5 = 0.18$, and the conditional complexity $K(\text{rider} | \text{horse}) = 2.47$ bits.

By contrast, $f(\text{rider} | \text{the}) = 71/13300 = 0.005 \approx f(\text{rider})$, and so $K(\text{rider} | \text{the}) \approx K(\text{rider})$.

This makes sense: in a context in which "horse" has been mentioned, "rider" is less complex than in a context in which "the" has been mentioned (which amounts to no context at all).

Question: Conditional complexity is asymmetrical. Can we define a genuine distance based on it?

Answer: Yes, and this is what Paul Vitányi and his colleagues did in 2004 by defining the "Normalized Information distance" (NID).

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